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e, it is described as a "developmental disorder" because symptoms generally appear in the first 3 years of life [1,2]. For many years, a diagnosis of autism was rare, occurring in just one child out of 10,000. However, since the mid-1980s, the rate of autism has increased dramatically around the world. The World Health Organization (WHO) estimates that one in 160 children in the world has ASD [3], while the prevalence in the world was 0.6% [4]. The prevalence was also significantly higher for children aged between 6 and 12 years compared to children under the age of 5 and over the age of 13 years [5].

Although there doesn't seem to be any oral manifestations specific to autism, certain conditions may appear due to behaviors related to autism, such as communication limitations, personal negligence, self-injurious behavior, eating habits, effects of medication, resistance to oral care, and hyposensitivity to pain [6]. Children with ASD are widely reported to have poorer oral hygiene and worse periodontal conditions, due to a lack of adequate manual skills for brushing teeth, but also due to a lack of interest in this regard [6-8]. Additionally, autistic children often prefer frequent, soft, and sweet foods and tend to keep food in their mouths longer than usual, which further contributes to the development of caries [9].

Proper oral care at home, including a balanced diet, regular brushing habits, and regular dental visits, is crucial for improving the oral health of children with autism. Studies show that children with autism often brush their teeth irregularly; in these cases, parental supervision is necessary [10,11].

Parents are the primary decision-makers in matters of healthcare for children with autism; thus, they play a crucial role in achieving the best oral health outcomes for their children. Since parents are primarily responsible for most health issues related to their children, their role in modeling preventive oral health practices throughout their children's lives is crucial [12].

Parents of children with autism are often physically, financially, and psychologically overburdened. These children depend mainly on their parents for their daily needs, as well as for oral healthcare. Consequently, one parent is often forced to stay home, which negatively impacts the family's financial situation [12,13].

Therefore, the present study aimed to assess the oral health status of children with autism, identify factors that may affect their oral health, and examine parental attitudes regarding the care of their children's oral health.

■ Material and Methods

Study Design and Population

This is a cross-sectional study in which a group of children diagnosed with autism was examined. Initially, the association for the daily care of children with autism was contacted, and subsequently, through this association, the parents of the children attending the association (n = 60) were informed. Fifty-five parents of children with autism agreed to participate in the research. The research would be conducted at the association in the morning, when the parents brought their children, so that the oral examination of the children could be carried out in their presence, and they could also respond to a structured questionnaire. All the children participated in a dental examination, while their parents completed questionnaires. After the oral examination of the children, each parent was provided with a written and verbal report on their child's dental condition and the recommended dental treatment. Inclusion criteria were parental consent, children aged 6 - 11 years, and diagnosed with autism according to competent specialists. The exclusion criteria were the presence of systemic

ed questionnaire was developed to assess parental demographics and attitudes toward their children's oral health. Four dental specialists evaluated the content validity of the questionnaire in terms of clarity and comprehensibility. To assess the validity of the questionnaire, it was completed by five parents of children with Autism Spectrum Disorder. The respondents were asked about the comprehensibility and intelligibility of the questions, and it was found that they considered the questionnaire acceptable in these respects.

The questionnaire consisted of two parts. The first part addressed the demographic characteristics of the parents with autistic children, including age group, educational level, employment status, and socioeconomic status.

The education level of parents was assessed on four levels, including primary school, secondary school, university, and postgraduate studies. The socioeconomic status was evaluated according to the minimum wage in Kosovo, which is considered to be 250 euros.

The second part comprised seven multiple-choice questions to evaluate the children's oral health behaviors as reported by their parents. This included the daily frequency of consumption of sweets, beverages, fruits, and vegetables, the frequency of brushing teeth with or without parental supervision, the use of fluoride toothpaste, and the regularity of dental visits.

Pilot Study and Calibration

The inter-examiner agreement was assessed between two pediatric dental specialist examiners (the inter-examiner reliability through Cohen's kappa test was 0.95). First, the two examiners reviewed the examination method. They examined five children with autism to apply this method. Accordingly, the 50 remaining children with autism were examined by both examiners separately but in the same room. For intra-examiner calibration, the examiner, a pediatric dentist, examined five children to evaluate the dmft/DMFT index (d/Decayed, m/Missing, and f/Filled Teeth) and the Simplified Oral Hygiene Index (OHI-S). The second assessment was held approximately 30 minutes after the first.

Several aspects were considered for the calibration of the examiners: they had to be specialists in pediatric and preventive dentistry; they should have experience in treating both healthy children and those with various diseases, including autism; and they should also possess knowledge acquired through research in the field of pediatric and preventive dentistry.

Data Collection

The examination of the children was conducted in a classroom at their daily care center, accompanied by their parents and teacher. A week prior, the examiners had filled the classroom with various photographs illustrating the oral examination process to prepare the children for examination day. The oral examination was conducted by two pediatric dentists from the University Dentistry Clinical Center of Kosovo (UDCCK), who sat in front of the child. The examination was done under natural light using a dental mirror and a probe.

For the caries assessments, all tooth surfaces were examined. Every visual change in enamel transparency during the early phases of demineralization was defined as a carious lesion and was recorded using

ified Oral Hygiene Index (OHI-S) was used to evaluate the oral hygiene of the autistic children. The plaque of the labial surface of the permanent maxillary right central incisor and the permanent mandibular left central incisor, the buccal surface of the permanent maxillary right first molar and the permanent maxillary left first molar, and the lingual surface of the permanent mandibular right first molar and the permanent mandibular left first molar were checked. The oral hygiene status of children was classified according to the OHI-S index rating as follows: good (score = 0-1.2), fair (score = 1.3-3), and poor (score = 3.1-6) [15].

Ethical Aspects

Written consent was obtained from the parents, and ethical approval was granted by the Association for the Daily Care of Children with Autism "AUTIZMI" in Pristina (Ref. No. 005/02.02.2024).

Statistical Analysis

Data analysis was performed using SPSS version 27 (SPSS Inc., IL, USA). Descriptive statistics were calculated, including the mean and standard deviations (SD) for the quantitative variables, and frequencies and percentages for categorical variables. Differences between outcome variables were compared using one-way ANOVA. The significance level was set at $p < 0.05$.

■ Results

The sample included participants ($N = 55$) between 6 and 11 years of age, of both genders. Out of 55 children with autism, 27 (40.09%) were from urban areas and 28 (50.90%) were from rural areas. While 31 (56.37%) children were male and 24 (43.63%) were female (Table 1).

Table 1. Children were analyzed based on gender and residence.

Residence	Gender				Total	
	Male		Female			
	N	%	N	%	N	%
Urban	17	54.80	10	41.70	27	40.09
Rural	14	45.20	14	58.30	28	50.90
Total	31	56.37	24	43.63	55	100.00

The mean and standard deviation of the dmft/DMFT index in children from urban areas were 4.22 ± 2.940 and 1.63 ± 1.305 , respectively. In contrast, the mean and standard deviation of the dmft/DMFT were found to be 3.86 ± 2.289 and 1.96 ± 1.815 , respectively, for children from rural areas. Furthermore, the d and D components of the dmft/DMFT index had higher values, exceeding those of all the other components (3.61 ± 2.520 and 1.80 ± 1.550). In total, the mean and standard deviation for dmft were higher (4.04 ± 2.641) compared to the mean and standard deviation for DMFT (1.79 ± 1.56) (Table 2). On the other hand, the overall OHI-S index was 1.67 ± 0.477 , slightly higher in rural areas (1.68 ± 0.616), compared to urban areas (1.67 ± 0.338) (Table 2).

		27	4.04±2.880	27	3.18±2.160	3.61±2.520
			0.15±0.530		0.68±1.510	0.42±1.020
			0.00±0.000		0.00±0.000	0.00±0.000
dmft			4.22±2.940		3.86±2.289	4.04±2.641
D	28		1.70±1.290	28	1.89±1.810	1.80±1.550
M			0.00±0.000		0.04±0.180	0.02±0.090
F			0.00±0.000		0.04±0.180	0.02±0.090
DMFT			1.63±1.305		1.96±1.815	1.79±1.560
OHI-S	55		1.67±0.338	55	1.68±0.616	1.67±0.477

The frequency distribution of the demographic characteristics of the parents is presented in Table 3. Of all the parents, 25 (45.5%) belonged to the age group of 31-40 years, 20 (36.4%) to the age group of 41-50 years, 7 (12.7%) to the age group of 18-30 years, while only 3 (5.5%) parents were over 50 years old. According to marital status, 50 (90.9%) of the parents were married, 4 (7.3%) were divorced, and 1 (1.8%) was single. Regarding parents' education, as shown in Table 3, a higher percentage of fathers had completed secondary school and university (63.6% and 23.6%), while most mothers had completed secondary and primary school (56.4% and 21.8%). On the other hand, almost all fathers were employed (98.2%), while the largest percentage of mothers was unemployed (78.2%). The middle socioeconomic status prevailed (70.9%), while the monthly income was low (50.9%) (Table 3).

Table 3. Demographic characteristics of the parents of autistic children.

Variables	N	%
Age Group (Years)		
18-30	7	12.7
31-40	25	45.5
41-50	20	36.4
50+	3	5.5
Marital Status		
Single	1	1.8
Married	50	90.9
Divorced	4	7.3
Father's Education		
Primary school	2	3.6
Secondary school	35	63.6
University	13	23.6
Postgraduate studies	5	9.1
Mother's Education		
Primary school	12	21.8
Secondary school	31	56.4
University	6	10.9
Postgraduate studies	6	10.9
Father's Employment		
Student	1	1.8
Unemployed	0	0.0
Employed	54	98.2
Mother's Employment		
Student	1	1.8
Unemployed	43	78.2
Employed	11	20.0
Socioeconomic Status		
Low	9	16.4

000 €	22	40.0
1000 €	5	9.1

Table 4 displays parents' knowledge and attitudes about their children's oral health. As shown, most parents reported that their children consume sweets and beverages (34.5% and 47.3%), as well as fruits and vegetables (43.6%), several times a day. Additionally, 28 (50.9%) parents reported that their autistic child brushed their teeth once a day, with 41.8% of them requiring parental supervision, and they almost always used fluoride toothpaste (69.1%). Concerning visits to the dentist, 83.6% of parents sent their children to the dentist only when necessary. Unfortunately, no children were sent to the dentist for routine visits (Table 4).

Table 4. Parents' knowledge and attitudes regarding their children's oral health.

Questionnaire	N	%
Does your child consume sweets?		
Several times a day	19	34.5
Once a day	18	32.7
Rarely	18	32.7
Does your child consume beverages?		
Several times a day	26	47.3
Once a day	11	20.0
Rarely	18	32.7
Does your child consume fruits and vegetables?		
Several times a day	24	43.6
Once a day	11	20.0
Rarely	20	36.4
Does your child brush their teeth?		
No	12	21.8
Yes, on their own	20	36.4
Yes, with parent/s' supervision	23	41.8
How many times a day does your child brush their teeth?		
Once a day	28	50.9
2-3 times a day	12	21.8
Several times a week	6	10.9
Never	9	16.4
Does your child use fluoride toothpaste?		
Always	38	69.1
Seldom	12	21.8
No idea	5	9.1
How often does your child have dental visits?		
Every 6 months	0	0.0
Once a year	9	16.4
Only when necessary	46	83.6

In comparing the dmft index, DMFT, OHI-S, and the demographic characteristics of the parents of autistic children using a one-way ANOVA test, we found significant differences in the dmft index across different age groups (Table 5). Additionally, the DMFT and OHI-S index values were significantly influenced by the father's level of education and socioeconomic status (Table 5).

The analysis of variation between dmft, DMFT, and OHI-S index scores and parents' attitudes toward their children's oral health was conducted using the one-way ANOVA test, as shown in Table 6. A significant difference in dmft, DMFT, and OHI-S index values is observed in children who often consume sweets and

on of dmft, DMFT, and OHI-S index and demographic characteristics of the parents
n.

	dmft		DMFT		OHI-S	
	Mean±SD	p-value	Mean±SD	p-value	Mean±SD	p-value
Age Group						
18-30	6.57±2.507	0.001	2.00±1.633	0.684	1.90±0.684	0.061
31-40	4.00±2.483		2.00±1.633		1.76±0.503	
41-50	2.70±1.867		1.45±1.538		1.45±0.333	
50+	7.33±1.155		2.00±1.732		1.94±0.485	
Marital Status						
Single	4.00±0.000	0.214	2.00±0.000	0.582	1.50±0.000	0.867
Married	3.86±2.515		1.86±1.629		1.68±0.517	
Divorced	6.25±3.501		1.00±0.816		1.58±0.165	
Father's Education						
Primary school	2.00±0.000	0.365	5.00±0.000	0.019	2.66±0.000	0.017
Secondary school	4.29±2.685		1.83±1.543		1.70±0.511	
University	3.31±2.323		1.46±1.450		1.50±0.372	
Postgraduate studies	5.00±3.001		1.20±1.095		1.60±0.279	
Mother's Education						
Primary school	3.42±2.644	0.324	2.08±1.443	0.148	1.86±0.300	0.406
Secondary school	4.55±2.815		1.74±1.570		1.62±0.532	
University	4.00±2.191		0.67±1.033		1.49±0.276	
Postgraduate studies	2.67±1.033		2.67±1.966		1.75±0.726	
Father's Employment						
Student	8.00±0.000	0.127	0.00±0.000	0.254	1.33±0.000	0.490
Unemployed	0.00±0.000		0.00±0.000		0.00±0.000	
Employed	3.96±2.577		1.83±1.575		1.68±0.498	
Mother's Employment						
Student	8.00±0.000	0.080	0.00±0.000	0.487	1.33±0.000	0.773
Unemployed	4.26±2.691		1.79±1.552		1.67±0.499	
Employed	2.82±1.722		2.00±1.732		1.71±0.515	
Socioeconomic Status						
Low	5.00±2.500	0.269	1.00±1.118	0.135	1.53±0.370	0.556
Middle	4.03±2.661		1.85±1.582		1.68±0.504	
High	2.86±2.268		2.57±1.813		1.80±0.602	
Monthly Family Income						
250-500 €	3.75±2.444	0.405	1.82±1.565	0.280	1.68±0.483	0.636
600-1000 €	4.59±2.806		1.55±1.438		1.62±0.467	
Over 1000 €	3.20±2.683		2.80±2.168		1.87±0.728	

Table 6. Comparison of dmft, DMFT, and OHI-S index and parental attitudes towards the oral health of their children with autism.

Variables	dmft		DMFT		OHI-S	
	Mean±SD	p-value	Mean±SD	p-value	Mean±SD	p-value
Does your child consume sweets?						
Several times a day	4.21±2.898	0.875	1.79±1.475	0.994	1.66±0.470	0.769
Once a day	3.78±2.130		1.83±1.790		1.61±0.546	
Rarely	4.11±2.847		1.78±1.555		1.73±0.488	
Does your child consume beverages?						
Several times a day	4.38±2.639	0.102	1.73±1.538	0.672	1.61±0.512	0.322
Once a day	4.91±2.914		2.18±1.471		1.87±0.477	
Rarely	3.00±2.142		1.67±1.749		1.62±0.475	
Does your child consume fruits and vegetables?						
Several times a day	4.71±2.476	0.026	1.38±1.469	0.123	1.57±0.502	0.427

	2.92±2.065	1.42±1.975	1.56±0.585			
week	5.67±2.944	2.00±1.265	1.74±0.493			
	3.78±1.481	1.56±1.59	1.51±0.528			
How often does your child have dental visits?						
Every 6 months	0.00±0.000	0.077	0.00±0.00	0.466	0.00±0.00	0.677
Once a year	5.44±3.245		1.44±1.667		1.73±0.616	
Only when necessary	3.76±2.415		1.87±1.572		1.66±0.475	

■ Discussion

Nowadays, autism is a significant public health concern for the world population. Taking care of the oral health of these children also requires understanding, patience, and a commitment to improving their oral health condition [16]. Studies show an increased risk of oral health diseases in children with autism, also reported by parents of autistic children, with an impact on the quality of life of the family and the child [17,18].

This study sheds light on the state of oral health of children with autism and the parental attitudes toward these children's oral health. According to the WHO criteria, the mean of the dmft/DMFT index among children with autism is considered to be of woefully high level [19]. These findings are in agreement with other studies that also reported a higher value of dental caries in autistic children [20–22]. However, this result differs from other studies that found children with autism had a lower dmft/DMFT index [7,10,23]. A high caries index in children with autism in our study can be attributed to their poor oral hygiene, as indicated by the high dental plaque index, regardless of residence. This finding was consistent with previous studies [24,25]. These results suggest an immediate improvement in parental attitudes towards dental health, with the intention of preventing dental caries in children with autism.

Educated parents can better understand the disease of autism and the importance of caring for children with autism. The findings of this current study clarified that most of the children's parents had a secondary education, and some had pursued university and postgraduate studies. Despite this, employment status differed from education level, with the highest percentage of mothers not working. This may be regarding the caring challenges for autistic children, since they may require constant solicitude and supervision [23]. At the same time, this damages the family economy, where the study shows that the socioeconomic status is middle and the family's monthly income is not enough to provide for them all, especially for children with autism.

Studies have shown that regular intake of carbohydrates increases dental caries [26, 24], a finding that is also evident in our research and is consistent with other studies [24, 27, 28]. Children who consume sweets and beverages at least once a day are more likely to develop dental caries. This can be explained by the fact that parents give sweets to children with autism, especially in cases where the management of a particular situation becomes problematic and the child is comforted by sweets [29]. Nevertheless, a diet high in sugar, combined with poor oral hygiene, inevitably leads to poorer dental health.

Furthermore, autistic children are more liable to oral health problems. Since these children have difficulties in communication, behavior, and social skills, this complicates their dental care. This is mainly manifested by difficulty of brushing, a procedure they often have to do under parental supervision [23,30]. According to our findings, most children brush their teeth at least once a day, typically using fluoride toothpaste, and primarily under parental supervision. These findings were in agreement with previous studies [24,31].

lated to the increased anxiety of children being exposed to a new environment and their them feel uncomfortable. Most children visited the dentist when necessary, which is consistent with other studies [26, 33, 34]. Moreover, a study done by Suhaib et al. shows that an alarming majority of children with autism had never been to a dentist [30]. A possible reason could be the difficulty in finding a dentist to treat the child, combined with the child's compromised cooperation [35].

Most of the studies of children with autism are conducted in small populations. This study is the first to be performed on the oral health of children with autism in Kosovo, providing a wealth of basic information. An advantage of this study was the classroom environment, which provided a more suitable setting for examinations, as the children were accustomed to it.

The limitation of this study was that it was being conducted in only one daycare center. Hence, the sample size was small, and the results are not representative of the entire population. Further studies are recommended to overcome these limitations.

■ Conclusion

Children with autism showed high rates of caries, poor oral hygiene, and an urgent need for dental treatment. The mother's employment remains a challenge due to the constant care of the child with autism. Furthermore, the consumption of sugars contributes to the development of dental caries, so dental health professionals can implement clinical strategies to help reduce the intake of added sugars in patients with autism. Oral health programs should be used that emphasize the application of preventive measures through regular dental visits, which is considered of special importance for children with autism. Additionally, we recommend developing educational seminars for dentists, parents, and caregivers to enhance their skills in treating children with these conditions. A crucial aspect of such a program is collaboration among therapists, neuropsychiatrists, and dentists.

■ Authors' Contributions

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■ Financial Support

None.

■ Conflict of Interest

The authors declare that they have no conflicts of interest.

■ Data Availability

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